Generating Functions

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Theory

A **power series** has the form

$$f(x) = \sum_{n\geq 0} a_n x^n.$$

Given a sequence (b_n), we can consider the polynomial $P(x) = \sum_{n\geq 0}^{d} b_n x^n$ for an integer d, or the series $f(x) = \sum_{n\geq 0} b^n x^n$. This is called the **generating function** of the sequence b_n.

Example 1. Given that a_k is the number of subsets of $\{1, ..., n\}$ that have size k, show that

$$(1+x)^n = \sum_{k=0}^{n} a_k x^k.$$

Some relevant results are the following:

- $\sum_{k\geq 0} x^k = \frac{1}{1-x}$
- $\sum_{k\geq 0} kx^k = \frac{x}{(1-x)^2}$
- $\sum_{k\geq 0} a^k x^k = \frac{1}{1-ax}$
- $\sum_{k\geq 0} \frac{x^k}{k!} = e^x$
- $\sum_{k\geq 0} \binom{r}{k} x^k = (1+x)^r$
- $\sum_{k\geq 0} \binom{n+k-1}{k} x^k = \frac{1}{(1-x)^n}$

Example 2. Find a closed-form formula for

$$\binom{n}{0} + \binom{n}{3} + \binom{n}{6} + ...$$

Problems

Problem 1. For a permutation π of $\{1, 2, ..., n\}$, let the number of inversions of π, denoted $inv(\pi)$, be the number of pairs (k, j) such that $k < j$ and $\pi(k) > \pi(j)$. Let S_n be the set of permutations with n elements.
Prove that

$$\sum_{\pi\in S_n} x^{inv(\pi)} = \prod_{j=0}^{n-1} (1 + x + ... + x^j).$$

Problem 2. A sequence (F_n) is defined as $F_0 = 0, F_1 = 1, F_{n+1} = F_n + F_{n-1}$, $\forall n \geq 1$. Find a closed-form expression of F_n.

Problem 3. Find the n-th term of the series that satisfies $a_n = 2a_{n-1} + 1$, $a_0 = 0$.

Problem 4. How many n-digits numbers, whose digits are in the set $\{2, 3, 7, 9\}$ have sum of digits divisible by 3?

Problem 5. Let $n \geq 2$ and $N = 2^n$. Let $A_1, A_2, ..., A_N$ be a permutation of all the subsets of $X = \{1, 2, ..., n\}$. Determine the maximum possible value of the sum:

$$S = \sum_{i=1}^{N} | A_i \cap A_{i+1} | \cdot | A_i \cup A_{i+1} |$$

where $A_{N+1} = A_1$.

Problem 6. Let p be an odd prime number. How many p-element subsets A of $\{1, 2, ..., 2p\}$ are there, the sum of whose elements is divisible by p?